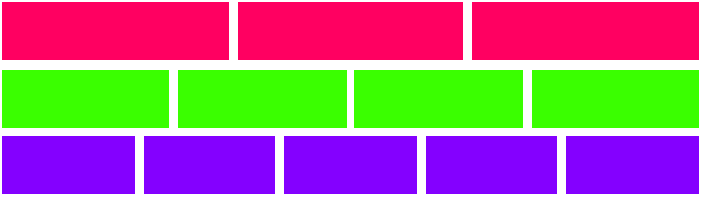
Polyrhythmic Design

Polyrhythmic Design

Rhythm, Harmony, Hue:
Music Theory You Can See

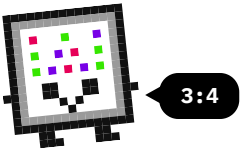
Steve Chab
//polyrhythmic.design



PART I

The idea

Three against four. A magazine grid. A chord.
Forty doublings from a drumbeat to a color.



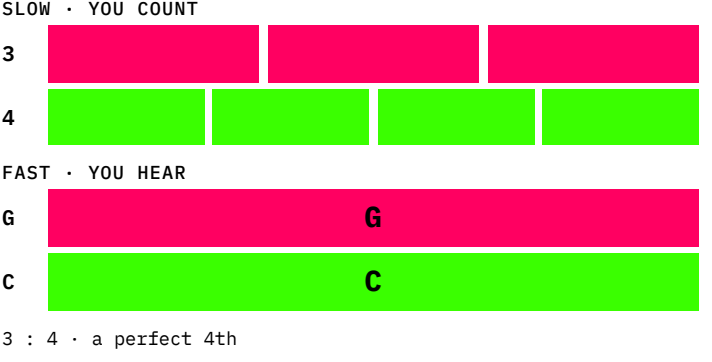
Poly, your guide.

Three against four

Turn a crank slowly and you hear clicks. Turn it faster and the clicks blur into a hum. A playing card in bicycle spokes does the same: tick, tick, tick, then a buzz. A truck engine idles in clicks then revs into a note. So does a fan.

Rhythm works that way too. Play three beats against four, slowly, and you hear a groove. Speed it up and your ear stops counting and starts hearing: a perfect 4th, two notes at once.

The ratio did not change. Only the clock did.



3 : 4 · a perfect 4th

Forty doublings

Double a note and you get the same note, one octave up. Double a drum beat eight times and you reach A, 440 Hz. Double that forty more times and you land near 484 trillion cycles a second: orange-red light.

Sound and light are different kinds of wave, though. Sound is air pushing on air. Light is an electric and magnetic ripple that crosses empty space. Doubling a drumbeat won't make it glow. What carries across is the ratio.

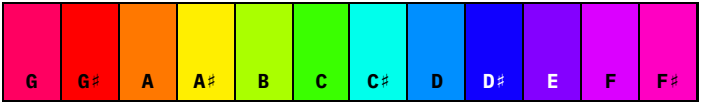
1.72 BEATS A SECOND

× 2^{8} ↓

440 HZ · A4, A NOTE

× 2^{40} ↓

483.8 THZ · 620 NM · A IS ORANGE-RED



The twelve notes as light, G to F♯. Visible light runs about 400 to 790 THz, just under one octave.

Reading a swatch



E
#8400ff
H271 S100 L50
C73 M77 Y0 K0

← Note name
← Hex code
← HSL, the model
← CMYK, a guide

Every color here is a note. Each comes in three vibrancies: Soft, Pure and Deep. Pure is the true color. Wavelength and frequency are on page 138.

SOFT	PURE	DEEP
#c280ff	#8400ff	#420080

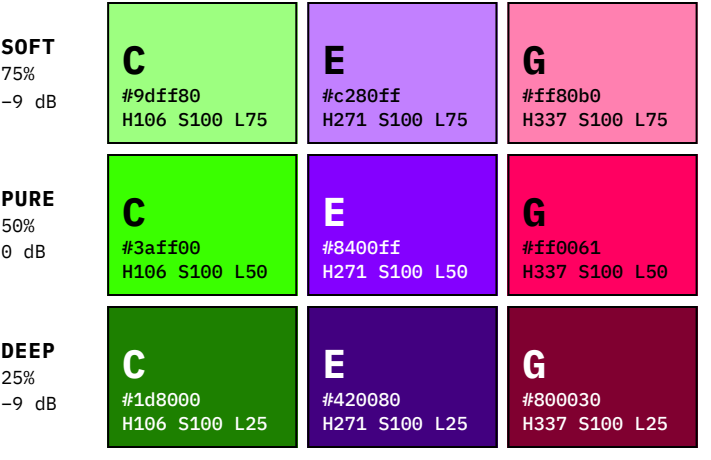
Screens add light. Paper subtracts it.

A press rebuilds every color from four inks, so the brightest greens, cyans and blues print duller than they look on a screen. The hex codes are exact. The CMYK numbers are a guide: the printer makes its own conversion.

One chord, three volumes

Lightness is volume. Pure is the true color at full volume. Soft is lighter and quieter. Deep is darker and just as quiet. At the ends, white and black are silent.

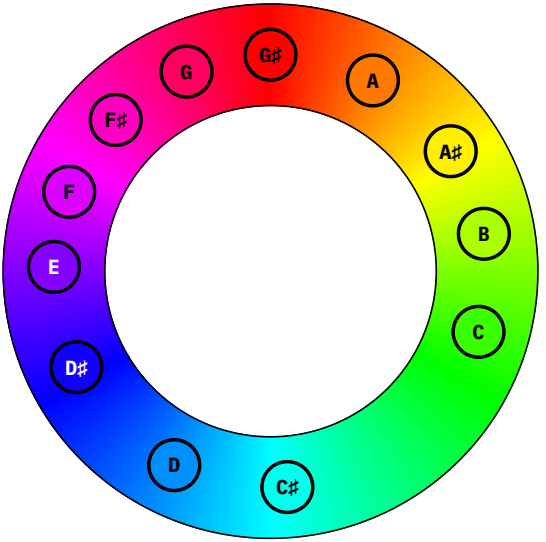
But white light is every hue at once. Shouldn't white be every note at once? In sound, every pitch at once is noise, and the apps keep noise on its own dial: saturation, from a pure tone down to white noise. That leaves lightness to mean volume alone. It's a convention, so each dial does one job.



C major · C E G

Traditional

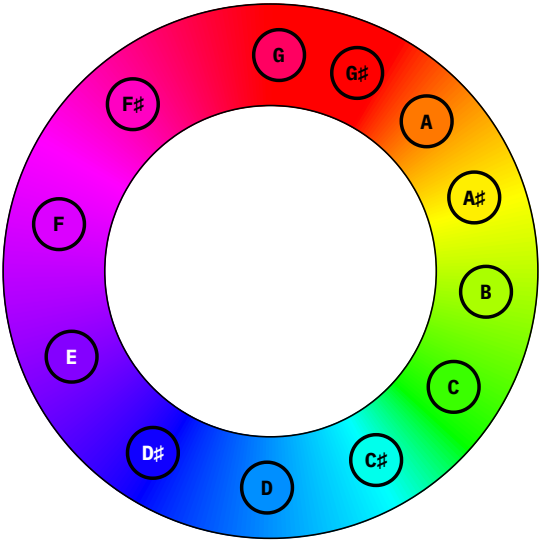
Every color family gets 60°.



Painters' wheels and screen wheels spread red, yellow, green, cyan, blue and magenta evenly. The twelve notes sit where their hues fall.

Spectrum

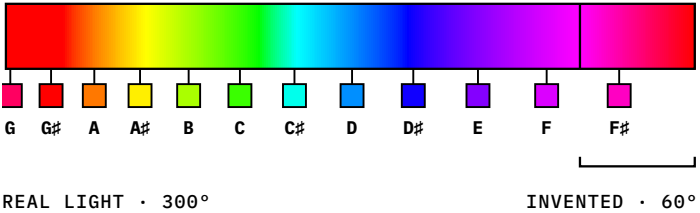
Spaced by frequency, terahertz by terahertz.



Same notes, spaced by frequency. Violet gets the most room, then red: light from about 620 to 750 nm all looks red. Green gets squeezed. Pinks have no wavelength, so F♯ and G borrow them (page 14).

Pink is not in the rainbow

Every color in a rainbow is one wavelength of light. Pink isn't. No wavelength makes it. Your eye sees pink when red light and violet light arrive together with no green between, and your brain, refusing to leave the color circle open, invents a bridge. Pink is the brain's own idea. A good one.



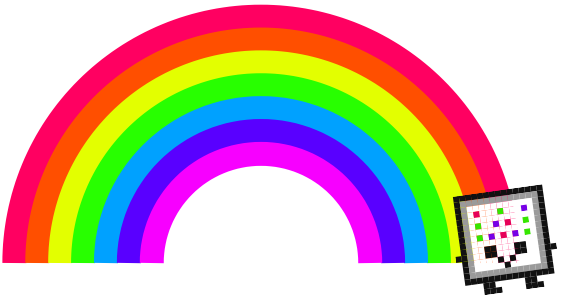
Ten of the twelve notes keep their true color. Two live on the bridge. F♯ is ultraviolet, 368 nm, a light no one can see, so it gets the invented pink. G is the same red as G♯, so it slides onto the bridge to keep the two apart.

The blend: one straight line of hue from F's violet (292°) to G♯'s red (360°), cut into three equal steps. F♯ lands at 314°, G at 337°.

Everyone's favorite made-up color

Blue wins most favorite-color polls. In 2015 YouGov asked people in ten countries, and blue came first in every one. Pink does better than the polls suggest: in a 2007 Newcastle University study, women's choices leaned from blue toward red, which made pinks and lilacs their most preferred colors.

Is pink loved because the brain invents it? No study we found says so. We like to think so anyway.



A 7-EDO rainbow: seven equal steps for Roy G. Biv's seven colors, starting on G's red and ending in violet.